

Notes: Hennessey and Whited (JF, 2005)

Debt Dynamics

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1 Big picture

- **Question.** Can a rational *dynamic* trade-off model explain the main leverage facts that seem to reject the *static* trade-off theory: debt conservatism, the negative relation between profitability and leverage, leverage hysteresis, and the Baker–Wurgler weighted- Q result?
- **Core puzzle.** In the static trade-off theory, profitable and liquid firms should borrow more because they have more income to shield and lower expected distress costs. In the data, many profitable firms borrow less, and leverage appears to depend on history.
- **Main idea.** The right comparison is not always *debt versus external equity*. Sometimes the relevant margin is *retention versus distribution*. Sometimes a current debt choice changes the probability that the firm will need equity tomorrow. Once the financing margin becomes endogenous, tax incentives interact with liquidity and future financing needs in a very different way.
- **Model environment.** The firm chooses leverage, saving, payouts, and real investment jointly in an infinite-horizon setting with a progressive corporate tax schedule, personal taxes on interest and distributions, equity flotation costs, financial distress costs, and a collateral constraint.
- **Central mechanism.** The marginal benefit of debt depends on what the dollar of debt is replacing. If extra debt replaces costly external equity, debt is attractive. If extra debt only finances higher shareholder distributions, debt is less attractive. Therefore the sign of tax effects depends on the firm’s current and anticipated equity regime.
- **Main theoretical result.** There is **no single target leverage ratio**. The optimal policy is regime-dependent and piecewise. Firms can be net savers or highly levered, and leverage is path dependent because current debt affects the current financing gap and therefore today’s margin.
- **Empirical implications.** The model implies that leverage should be decreasing in lagged liquidity, that leverage should display hysteresis, and that current leverage can be negatively related to an external-finance-weighted average Q even without market timing or adverse-selection premia.
- **Simulation result.** A benchmark simulation with small flotation costs already reproduces the main stylized facts: negative leverage–cash-flow coefficients, negative leverage–EBITDA coefficients, and a negative coefficient on weighted Q .
- **Structural-estimation result.** Using indirect inference on Compustat data, the paper estimates a modest external-equity cost of about 5.9%. The authors argue that taxes and dynamic financing margins, more than large adverse-selection premia, explain the empirical anomalies.
- **Takeaway.** Much of the empirical evidence against the static trade-off theory is not evidence against tax-based optimizing behavior. It is evidence that financing is a dynamic problem, not a one-shot date-zero problem.

2 Data and measurement (key objects)

2.1 The firm in the model

The state at the beginning of a period is (k, p, z) :

- k : installed capital.
- p : inherited one-period face-value debt; negative p means the firm is a net saver.
- z : profitability or productivity shock.

The firm chooses next period's capital and debt, (k', p') , and the implied payout or equity issuance.

The operating-profit function is

$$\pi(k, z), \tag{1}$$

which is increasing and concave in k . In the simulations, the authors use

$$\pi(k, z) = zk^\alpha, \tag{2}$$

with $\alpha < 1$.

2.2 Taxes and financing frictions

The key tax parameters are:

- τ_i : personal tax rate on interest income.
- τ_d : personal tax rate on corporate distributions.
- $g(y)$: corporate tax bill as a convex function of taxable income y .
- $\tau_c(y) = g_1(y)$: marginal corporate tax rate.

The key financing frictions are:

- $\lambda > 0$: proportional flotation cost of external equity.
- $s < 1$: fire-sale value of capital in distress.
- a collateral constraint that caps debt using the fire-sale value of assets.

The paper's important conceptual point is that the tax advantage of debt is *not* constant. It varies with current liquidity, expected future financing needs, and the firm's location in one of three equity regimes:

- positive distributions,
- zero distributions,
- external equity issuance.

2.3 Key empirical sample

The structural estimation uses the full-coverage 2002 Compustat industrial files. The final estimation sample:

- covers **1993–2001**,
- excludes regulated and financial firms,
- excludes observations with missing values, nonpositive assets/capital/sales, small firms subject to rounding issues, and observations that fail basic accounting identities,
- yields an unbalanced panel with roughly **592 to 1128 firm-year observations per year**.

The sample starts in 1993 because the tax parameters the authors want to model are meant to be relevant for that period.

2.4 Main variables in the simulations and the data

The paper constructs model counterparts to standard empirical variables:

- **Investment/assets:**

$$\frac{k' - (1 - \delta)k}{k}.$$

- **Cash flow/assets:**

$$\frac{\pi(k, z) - g[y(k, p, z)] - p}{k}.$$

- **Tobin's Q :**

$$\frac{V(k, p, z) + p'/(1 + r)}{k}.$$

- **Debt/market assets:**

$$\frac{p'/(1 + r)}{V(k, p, z) + p'/(1 + r)}.$$

- **EBITDA/assets:**

$$\frac{\pi(k, z)}{k}.$$

- **Equity issuance/assets:**

$$\frac{-e(k, p, k', p', z)}{k}$$

whenever the distribution to equity is negative.

In the Compustat data used for estimation:

- assets are item #6,
- investment is item #30,
- equity issuance is item #108 minus item #115,
- net debt is item #9 plus item #34 minus item #1,
- operating income is item #13 divided by item #6.

2.5 Moments used for structural estimation

The indirect-inference estimator matches **10 moments**:

1. average investment/assets,
2. variance of investment/assets,
3. average EBITDA/assets,
4. average debt/assets ratio net of cash,
5. average equity issuance/assets,
6. frequency of equity issuance,
7. investment- Q sensitivity,
8. debt- Q sensitivity,
9. serial correlation of income/assets,
10. standard deviation of the shock to income/assets.

These moments are intended to identify the curvature of profits, depreciation, financing costs, distress costs, and the stochastic process for profitability.

3 Models and empirical specifications (all equations + interpretation)

3.1 Static trigger conditions from the paper's basic argument

The paper starts with simple one-period examples to show why the static Miller-style trigger is too narrow.

If a date-zero firm compares *debt versus external equity*, then with zero flotation costs the standard condition for debt dominance is

$$\tau_c > \frac{\tau_i - \tau_d}{1 - \tau_d}. \quad (3)$$

Interpretation. This is the familiar Miller trigger. It is valid only when the firm is at the debt-versus-equity margin and expects the marginal dollar next period to be distributed.

If instead the firm expects *external equity to be the marginal source of funds next period*, then the relevant comparison becomes

$$\tau_c > \tau_i. \quad (4)$$

If the firm is flush with internal funds and the true margin is *retention versus distribution*, the same condition reappears:

$$\tau_c > \tau_i. \quad (5)$$

Interpretation. The critical tax rate depends on the financing margin. This is the paper's first big

message: the “target tax rate” is regime-dependent, so the static target in equation (3) is generally not the right benchmark.

3.2 Taxable income and the corporate tax schedule

Corporate taxable income is

$$y(k, p, z) = \pi(k, z) - \delta k - r \left(\frac{p}{1+r} \right), \quad (6)$$

and the marginal corporate tax rate is

$$\tau_c[y(k, p, z)] = g_1[y(k, p, z)]. \quad (7)$$

Interpretation.

- Higher current debt lowers taxable income through interest deductions.
- Because $g(\cdot)$ is convex, the effective tax benefit of extra debt depends on the full distribution of future taxable income, not just on today’s income.
- This convexity is the tractable stand-in for the real-world asymmetry created by loss carrybacks and carryforwards.

3.3 Collateral constraint and fire-sale technology

The lender requires the fire-sale value of capital to cover promised repayment:

$$p' \leq sk'(1 - \delta) + \pi(k', z') - g[y(k', p', z')]. \quad (8)$$

If after-tax internal funds are not enough to cover the debt payment, the firm sells assets in a fire sale:

$$n(k', p', z') = \max \left\{ 0, \frac{p' - [\pi(k', z') - g(y(k', p', z'))]}{s} \right\}. \quad (9)$$

The funds needed to move the capital stock to k' are

$$i(k, p, k', z) = k' - [k(1 - \delta) - n(k', p', z')]. \quad (10)$$

Interpretation.

- Distress is costly because a dollar of repayment may require selling more than a dollar of productive capital.
- The collateral constraint prevents the model from choosing implausibly large debt positions and keeps the numerical problem tractable.

3.4 Net cash flow to equity

Pre-financing cash flow is the firm's internal funds net of taxes and debt service, plus new borrowing minus investment. The paper writes net cash flow to shareholders as

$$e(k, p, k', p', z) = [1 + \mathbf{1}_i \lambda - \mathbf{1}_d \tau_d] \left[\frac{\pi(k, z) - g[y(k, p, z)] - p}{\mathbf{1}_n + s \mathbf{1}_s} - (k' - k(1 - \delta)) + \frac{p'}{1 + r} \right], \quad (11)$$

where:

- $\mathbf{1}_i = 1$ in the equity-issuance regime,
- $\mathbf{1}_d = 1$ in the positive-distribution regime,
- $\mathbf{1}_n$ and $\mathbf{1}_s$ indicate no-fire-sale and fire-sale states.

Interpretation.

- If $e < 0$, the firm issues equity and pays the flotation cost.
- If $e > 0$, the firm distributes funds and shareholders pay the distribution tax.
- If $e = 0$, the firm sits in the interior zero-distribution regime.

3.5 The Bellman equation

The manager maximizes the discounted value of net cash flow to shareholders:

$$V(k, p, z) = \max_{(k', p') \in K \times P} \left\{ e(k, p, k', p', z) + \frac{1}{1 + r(1 - \tau_i)} \int V(k', p', z') \Delta(z, dz') \right\}. \quad (12)$$

The paper proves:

- existence and uniqueness of the value function,
- concavity of V in (k, p) ,
- a continuous single-valued optimal policy correspondence,
- differentiability away from the kink points created by fire sales and regime switches.

Interpretation. This is a standard dynamic programming problem, but the financing frictions create piecewise linear or kinked marginal conditions. Those kinks are the source of the regime structure.

3.6 Funding gap, marginal benefit, and marginal cost of debt

The budget constraint can be rearranged into a **funding-gap** form:

$$\frac{p'}{1 + r} - \frac{e(k, p, k', p', z)}{1 + \mathbf{1}_i \lambda - \mathbf{1}_d \tau_d} = i(k, p, k', z) - \max\{\pi(k, z) - g[y(k, p, z)] - p, 0\}. \quad (13)$$

Holding the right-hand side fixed and changing p' gives the iso-funding derivative

$$\left(\frac{\partial e}{\partial p'}\right)_{\text{isofund}} = \frac{1 + \mathbf{1}_i \lambda - \mathbf{1}_d \tau_d}{1 + r}. \quad (14)$$

The paper's central decomposition is

$$(1 + r)\Delta(k, p, k', p', z) = \text{MB}(k, p, k', p', z) - \text{MC}(k', p', z), \quad (15)$$

with

$$\text{MB}(k, p, k', p', z) = 1 + \mathbf{1}_i \lambda - \mathbf{1}_d \tau_d, \quad (16)$$

and

$$\text{MC}(k', p', z) = \int \frac{[1 + \mathbf{1}_i \lambda - \mathbf{1}_d \tau_d + \mathbf{1}_0 \phi][1 + r(1 - \tau_c(y(k', p', z')))]}{[1 + r(1 - \tau_i)](\mathbf{1}_n + s\mathbf{1}_s)} \Delta(z, dz'). \quad (17)$$

Interpretation.

- The marginal benefit of debt is *high*, $1 + \lambda$, when debt substitutes for costly external equity.
- The marginal benefit of debt is *low*, $1 - \tau_d$, when extra debt only finances a higher payout to shareholders.
- The marginal cost of debt depends on expected future corporate tax rates, expected distress, and expected future equity regime.

3.7 The regime-switch point and why there is no target leverage ratio

Define the switch point p'_0 by

$$e(k, p, k', p'_0, z) = 0. \quad (18)$$

Equivalently,

$$\frac{p'_0(k, p, k', z)}{1 + r} = i(k, p, k', z) - \max\{\pi(k, z) - g[y(k, p, z)] - p, 0\}. \quad (19)$$

The optimal financing policy has three cases:

$$\frac{p'_0}{1 + r} > H_i \Rightarrow \frac{p'}{1 + r} = H_i, \quad e < 0, \quad (20)$$

$$\frac{p'_0}{1 + r} < L_i \Rightarrow \frac{p'}{1 + r} = L_i, \quad e > 0, \quad (21)$$

$$\frac{p'_0}{1 + r} \in [L_i, H_i] \Rightarrow p' = p'_0, \quad e = 0. \quad (22)$$

Interpretation.

- If the funding gap is very large, the firm uses debt up to a threshold and then issues equity.
- If the funding gap is very negative, the firm borrows enough to make distributions.

- In the middle, it chooses debt exactly so that payouts are zero.

This piecewise rule is why there is **no universal target leverage ratio**. Optimal leverage depends on the current state and on which side of the regime boundaries the firm lies.

3.8 Three marginal-cost scenarios

Figure 1 in the paper summarizes three qualitatively different cases:

$$\text{MC}(k', 0, z) < 1 - \tau_d \quad (\text{low MC}_1 \text{ case}), \quad (23)$$

$$1 - \tau_d < \text{MC}(k', 0, z) < 1 + \lambda \quad (\text{intermediate MC}_2 \text{ case}), \quad (24)$$

$$\text{MC}(k', 0, z) > 1 + \lambda \quad (\text{high MC}_3 \text{ case}). \quad (25)$$

Interpretation.

- **Low MC₁.** Debt is so attractive that even unconstrained firms may issue debt and distribute cash.
- **Intermediate MC₂.** The firm behaves roughly like a pecking-order firm: unconstrained firms save or distribute, very constrained firms combine debt and equity.
- **High MC₃.** Debt is unattractive; some firms save and issue equity at the same time.

These cases show why the dynamic model can generate both debt conservatism and apparently strange joint behavior such as saving plus equity issuance.

3.9 Optimal investment

When the financing policy adjusts optimally, the investment first-order condition becomes

$$1 + \mathbf{1}_i \lambda - \mathbf{1}_d \tau_d = \frac{1 + \mathbf{1}_i \lambda - \mathbf{1}_d \tau_d + \mathbf{1}_0 \hat{\phi}}{1 + r(1 - \tau_i)} \int \left[\frac{\pi_1(k', z')(1 - \tau_c) + \delta \tau_c}{\mathbf{1}_n + s \mathbf{1}_s} + (1 - \delta) \right] \Delta(z, dz'). \quad (26)$$

Interpretation.

- The marginal cost of investment to shareholders depends on the current financing regime.
- Investment is more attractive when retained funds are cheap and less attractive when the marginal dollar must come from external equity.
- This is why financing and investment must be solved jointly rather than sequentially.

3.10 Simulation design and auxiliary regressions

For simulation, the profitability shock follows

$$\ln z' = \rho \ln z + \varepsilon', \quad \varepsilon' \sim N(0, \sigma_\varepsilon^2). \quad (27)$$

The simulated cross section is evaluated with the same kinds of regressions used in the empirical literature:

$$\frac{\text{Debt}}{\text{Market Assets}} = a_0 + a_1(\text{Tobin's } Q) + a_2 \left(\frac{\text{Cash Flow}}{\text{Book Assets}} \right) + u_a, \quad (28)$$

$$\frac{\text{Debt}}{\text{Market Assets}} = b_0 + b_1(\text{Tobin's } Q) + b_2 \left(\frac{\text{EBITDA}}{\text{Book Assets}} \right) + u_b, \quad (29)$$

$$\frac{\text{Debt}}{\text{Market Assets}} = c_0 + c_1(\text{Tobin's } Q) + c_2(\text{Weighted } Q) + c_3 \left(\frac{\text{EBITDA}}{\text{Book Assets}} \right) + u_c. \quad (30)$$

Interpretation. The paper does not claim these regressions are structural. They are diagnostic objects used to ask whether the dynamic trade-off model can generate the signs seen in reduced-form capital-structure regressions.

4 Identification and estimation strategy (what makes the exercise credible)

4.1 Why indirect inference is used

The model has discrete financing regimes and kinked policies, so there is no smooth closed-form Euler equation or decision rule that can be estimated directly by standard regression methods. The authors therefore estimate the structural parameter vector

$$b = (\alpha, \delta, \lambda, s, \sigma_\varepsilon, \rho, \mu_\tau, \sigma_\tau) \quad (31)$$

by matching simulated moments to sample moments.

4.2 The estimator

Let $\widehat{M}_N$ denote the vector of sample moments from the actual data, and let $\widehat{m}_N^s(b)$ denote the corresponding moments computed from simulated panel s under parameter vector b . The estimator is

$$\widehat{b} = \arg \min_b \widehat{G}_N' \widehat{W}_N \widehat{G}_N, \quad \widehat{G}_N = \widehat{M}_N - \frac{1}{S} \sum_{s=1}^S \widehat{m}_N^s(b). \quad (32)$$

The weighting matrix $\widehat{W}_N$ is the inverse asymptotic covariance matrix of the sample moments, estimated using influence functions.

Interpretation. The model is asked to reproduce the empirical patterns in the data. Parameters are chosen not because a single equation fits well, but because the whole model can jointly account for multiple moments.

4.3 Simulation protocol

To make the simulated data comparable to Compustat, the paper:

- creates $S = 6$ simulated panels,
- simulates **10,000 i.i.d. firms** in each panel,
- runs each firm for **50 periods**,
- keeps only the **last 9 periods** to match the data span,
- removes firm and year heterogeneity from the actual data using fixed effects before constructing moments.

Why this matters. The procedure is designed to compare like with like. The model produces homogeneous firms hit by shocks, so the empirical moments are purged of fixed heterogeneity before matching.

4.4 What is identified by which moments?

The moment choice has a clear logic:

- average investment and its variance help identify δ and the curvature of profits,
- average EBITDA/assets helps identify α and the profitability process,
- average debt, average equity issuance, and the frequency of issuance help identify λ and the distress/collateral parameters,
- the serial correlation and innovation variance of income identify $(\rho, \sigma_\varepsilon)$,
- the investment- Q and debt- Q sensitivities discipline the dynamic mapping from shocks into financing and investment.

The paper also emphasizes an informal local-identification check: if the matched moments barely change when parameters change, the Jacobian would be near singular and the standard errors would explode.

4.5 Why the authors claim the approach improves on simple regressions

The point is not that indirect inference is magic. The point is that it avoids a common simultaneity problem in reduced-form financing regressions.

In standard leverage regressions, debt, investment, profitability, and Q are all jointly determined. In this paper, the model chooses them simultaneously, and the estimator matches joint outcomes. The identifying restrictions come from the dynamic structure, not from a claim that a particular regressor is exogenous.

4.6 What makes the exercise convincing

Several features make the empirical exercise more persuasive:

- The model matches many moments at once rather than one regression coefficient.

- The overidentifying-restrictions statistic is modest:

$$\chi^2 = 4.906 \quad (p = 0.086).$$

This is not rejected at the 5% level.

- The model also delivers important *out-of-sample* patterns that were *not* directly used as matched moments: negative leverage sensitivities to cash flow, EBITDA, and weighted Q .

4.7 Limitations

The credibility of the exercise should be judged together with its limitations:

- It is a **partial-equilibrium** model; the interest rate and tax environment are taken as given.
- Debt is **single-period**, so the model abstracts from maturity choice.
- There are **no adjustment costs**, market timing motives, or adverse-selection premia built in.
- The tax schedule is approximated in reduced form.
- Simulated firms are i.i.d., so real-world heterogeneity is absorbed by fixed-effects removal rather than modeled directly.

These are meaningful abstractions, but they are deliberate: the paper wants to show how far a relatively parsimonious dynamic trade-off model can go.

5 Tables and figures

5.1 Figure 1: optimal financial policy

Read. Figure 1 draws three marginal-cost schedules, MC_1 , MC_2 , and MC_3 , together with two horizontal marginal-benefit lines:

- $1 + \lambda$ when extra debt substitutes for external equity,
- $1 - \tau_d$ when extra debt finances larger shareholder distributions.

The thresholds L_i and H_i determine whether the firm distributes, stays at zero payout, or issues equity.

Economic message. The figure is the cleanest picture of why there is no single target leverage ratio. Leverage depends on the firm’s current funding gap and on where the marginal-cost schedule sits. A shift in expected future taxes or expected future financing needs moves the MC schedule and changes the optimal leverage policy.

5.2 Table I: simple benchmark simulation

Read. In the benchmark simulation, the model produces the following main summary statistics:

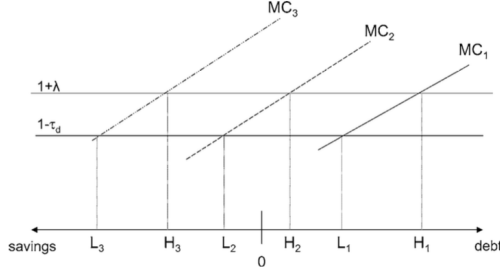


Figure 1. Optimal financial policy. This figure illustrates optimal financing in our model. The three MC lines represent the discounted marginal cost of debt service, taking expectations over the marginal corporate tax rate, distress costs, and the anticipated equity margin next period. At one extreme, the lowest schedule (MC_1) represents marginal debt service costs for a firm expecting a high corporate tax rate and to be making a distribution to shareholders next period. At the opposite extreme, the high schedule (MC_3) represents the marginal cost of debt service for a firm expecting a low corporate tax rate next period and anticipating a high probability of tapping external equity in that period. The MC_2 schedule represents an intermediate case. The horizontal lines represent the marginal benefit of debt issuance, which depends on the use of funds. When the increase in debt (reduction in savings) is used as a substitute for external equity, the line labeled $1 + \lambda$ measures the marginal benefit. When the increase in debt (reduction in savings) is used to finance an increase in distributions, the line labeled $1 - \tau_d$ measures the marginal benefit. The optimal financing policy equates the marginal costs and benefits of debt issuance.

Table I
Simple Model Simulation

All calculations are from a simulation of the dynamic partial-equilibrium model in Section II, which characterizes the firm's optimal choice of investment and capital structure in the face of corporate and personal taxes and costs of financial distress. The model is solved by value-function iteration and is simulated for 10,000 time periods, where the first 50 are dropped. ($a_1, a_2, b_1, b_2, c_1, c_2, c_3$) are estimated slope coefficients in the following regressions:

$$\frac{\text{Debt}}{\text{Market Assets}} = a_0 + a_1 (\text{Tobin's } Q) + a_2 \left(\frac{\text{Cash Flow}}{\text{Book Assets}} \right) + u_a$$

$$\frac{\text{Debt}}{\text{Market Assets}} = b_0 + b_1 (\text{Tobin's } Q) + b_2 \left(\frac{\text{EBITDA}}{\text{Book Assets}} \right) + u_b$$

$$\frac{\text{Debt}}{\text{Market Assets}} = c_0 + c_1 (\text{Tobin's } Q) + c_2 (\text{Weighted } Q) + c_3 \left(\frac{\text{EBITDA}}{\text{Book Assets}} \right) + u_c$$

Average debt-assets ratio (net of cash)	0.095
Fraction of observations with positive cash	0.285
Average investment/assets	0.160
Standard deviation of investment/assets	0.169
Average equity issuance/assets	0.055
Frequency of equity issuance	0.139
a_1	-0.069
a_2	-0.865
b_1	-0.298
b_2	-1.378
c_1	-0.293
c_2	-0.908
c_3	-1.359

Statistic	Value
Average debt/assets ratio (net of cash)	0.095
Fraction of observations with positive cash	0.285
Average investment/assets	0.160
Frequency of equity issuance	0.139
Average equity issuance/assets (conditional on issuing)	0.055

The key reduced-form leverage coefficients are also all negative:

Coefficient	Estimate
a_1 on Tobin's Q	-0.069
a_2 on cash flow/assets	-0.865
b_1 on Tobin's Q	-0.298
b_2 on EBITDA/assets	-1.378
c_2 on weighted Q	-0.908

Economic message. Even with a small flotation cost ($\lambda = 0.028$ in the benchmark calibration), the model generates cash holdings, debt conservatism, and negative leverage–profitability and leverage–weighted- Q relations. This is the first direct attack on the idea that these signs automatically imply pecking order or market timing.

5.3 Table II: actual versus simulated moments

Read. The paper compares the 10 matched moments from Compustat with the model's simulated counterparts. Selected examples are:

Moment	Actual	Simulated
Average investment/assets	0.079	0.107
Average EBITDA/assets	0.146	0.165
Average debt/assets ratio (net of cash)	0.075	0.127
Average equity issuance/assets	0.042	0.052
Frequency of equity issuance	0.099	0.091
Investment- Q sensitivity	0.019	0.036
Debt- Q sensitivity	-0.080	-0.219
Serial correlation of income/assets	0.583	0.598
Shock s.d. of income/assets	0.117	0.120

Economic message. The fit is reasonably good on averages, issuance frequency, and shock dynamics. The biggest misses are the variance of investment and the Q sensitivities. The authors interpret the investment-variance miss as evidence that the real world has adjustment frictions not present in the model, and the Q miss as possible measurement noise in actual Q .

Table II
Simulated Moments Estimation: Moment Estimates

Calculations are based on a sample of nonfinancial firms from the annual 2002 COMPUSTAT industrial files. The sample period is 1993 to 2001. Estimation is done with the simulated moments estimator in Gourieroux et al. (1993), which chooses structural model parameters by matching the moments from a simulated panel of firms to the corresponding moments from these data. The simulated panel of firms is generated from the dynamic partial-equilibrium model in Section II, which characterizes the firm's optimal choice of investment and capital structure in the face of corporate and personal taxes and costs of financial distress. The model is solved by value-function iteration. The simulated panel contains 10,000 firms over 50 time periods, where only the last nine time periods are kept for each firm. This table reports the simulated and estimated moments.

	Actual Moments	Simulated Moments
Average investment/assets	0.079	0.107
Variance of investment/assets	0.006	0.014
Average EBITDA/assets	0.146	0.165
Average debt-assets ratio (net of cash)	0.075	0.127
Average equity issuance/assets	0.042	0.052
Frequency of equity issuance	0.099	0.091
Investment- q sensitivity	0.019	0.036
Debt- q sensitivity	-0.080	-0.219
Serial correlation of income/assets	0.583	0.598
Standard deviation of the shock to incomes/assets	0.117	0.120

Table III
Simulated Moments Estimation: Structural Parameter Estimates

Calculations are based on a sample of nonfinancial firms from the annual 2002 COMPUSTAT industrial files. The sample period is 1993–2001. Estimation is done with the simulated moments estimator in Gourieroux et al. (1993), which chooses structural model parameters by matching the moments from a simulated panel of firms to the corresponding moments from these data. The simulated panel of firms is generated from the dynamic partial-equilibrium model in Section II, which characterizes the firm's optimal choice of investment and capital structure in the face of corporate and personal taxes and costs of financial distress. The model is solved by value-function iteration. The simulated panel contains 10,000 firms over 50 time periods, where only the last nine time periods are kept for each firm. This table reports the estimated structural parameters, with standard errors in parentheses. α is the curvature parameter in operating profits, zK^α ; δ is the rate of capital stock depreciation; λ is the proportional cost of external equity; s is the resale price per dollar of used capital goods; σ_ε is the standard deviation of the innovation to $\ln(x)$; ρ is the serial correlation of $\ln(x)$; and μ_τ and σ_τ are parameters that define the shape of the marginal corporate tax schedule. χ^2 is a chi-squared statistic for the test of the overidentifying restrictions. In parentheses is its p -value.

α	δ	λ	s	σ_ε	ρ	μ_τ	σ_τ	χ^2
0.551 (0.276)	0.100 (0.059)	0.059 (0.028)	0.592 (0.352)	0.123 (0.106)	0.740 (0.262)	-12.267 (7.851)	9.246 (4.594)	4.906 (0.086)

5.4 Table III: structural parameter estimates

Read. The main parameter estimates are:

α	δ	λ	s	σ_ε	ρ	μ_τ	σ_τ
0.551	0.100	0.059	0.592	0.123	0.740	-12.267	9.246

The overidentifying-restrictions test is

$$\chi^2 = 4.906 \quad (p = 0.086).$$

Economic message.

- $\alpha = 0.551$ implies decreasing returns to scale.
- $\delta = 0.100$ is a plausible depreciation rate.

- $\lambda = 0.059$ says the cost of external equity is positive but modest.
- $s = 0.592$ implies meaningful but imprecisely estimated distress-sale discounts.
- The estimated tax-schedule parameters imply the firm sits at the top marginal corporate tax rate much of the time.

The authors' main empirical conclusion is that a large adverse-selection premium is *not* needed to explain the leverage facts.

5.5 Table III continuation: out-of-sample regression patterns

Read. Using the structural estimates from Table III, the model still implies:

$$\begin{aligned}\frac{\text{Debt}}{\text{Market Assets}} &= -0.045(\text{Tobin's } Q) - 0.754 \left(\frac{\text{Cash Flow}}{\text{Book Assets}} \right), \\ \frac{\text{Debt}}{\text{Market Assets}} &= -0.198(\text{Tobin's } Q) - 1.054 \left(\frac{\text{EBITDA}}{\text{Book Assets}} \right), \\ \frac{\text{Debt}}{\text{Market Assets}} &= -0.191(\text{Tobin's } Q) - 0.841(\text{Weighted } Q) - 1.042 \left(\frac{\text{EBITDA}}{\text{Book Assets}} \right).\end{aligned}$$

Economic message. These are the paper's strongest empirical moments because they are *not* directly the moments used to fit the model. The signs still line up with the well-known empirical facts.

Table III
Simulated Moments Estimation: Structural Parameter Estimates

Calculations are based on a sample of nonfinancial firms from the annual 2002 COMPUSTAT industrial files. The sample period is 1993–2001. Estimation is done with the simulated moments estimator in Gourieroux et al. (1993), which chooses structural model parameters by matching the moments from a simulated panel of firms to the corresponding moments from these data. The simulated panel of firms is generated from the dynamic partial-equilibrium model in Section II, which characterizes the firm's optimal choice of investment and capital structure in the face of corporate and personal taxes and costs of financial distress. The model is solved by value-function iteration. The simulated panel contains 10,000 firms over 50 time periods, where only the last nine time periods are kept for each firm. This table reports the estimated structural parameters, with standard errors in parentheses. α is the curvature parameter in operating profits, zK^α ; δ is the rate of capital stock depreciation; λ is the proportional cost of external equity; s is the resale price per dollar of used capital goods; σ_ε is the standard deviation of the innovation to $\ln(z)$; ρ is the serial correlation of $\ln(z)$; and μ_τ and σ_τ are parameters that define the shape of the marginal corporate tax schedule. χ^2 is a chi-squared statistic for the test of the overidentifying restrictions. In parentheses is its p -value.

α	δ	λ	s	σ_ε	ρ	μ_τ	σ_τ	χ^2
0.551 (0.276)	0.100 (0.059)	0.059 (0.028)	0.592 (0.352)	0.123 (0.106)	0.740 (0.262)	-12.267 (7.851)	9.246 (4.594)	4.906 (0.086)

Table IV
Simulation Summary Statistics

All calculations are from a simulation of the dynamic partial-equilibrium model in Section II, which characterizes the firm's optimal choice of investment and capital structure in the face of corporate and personal taxes and costs of financial distress. The model is solved by value-function iteration and is simulated for 10,000 time periods, where the first 100 are dropped. The parameterization for the first panel is given in Table III. The parameterization for the second panel is identical, except that λ is set to 0.

	Saving			Debt		
	Distributions	Equity	Neither	Distributions	Equity	Neither
$\lambda = 0.059$						
Fraction	0.199	0	0.086	0	0.091	0.624
Investment/assets	-0.032	-	-0.017	-	0.292	0.142
Debt/assets	-0.086	-	-0.064	-	0.286	0.198
Tobin's Q	1.945	-	1.806	-	2.846	2.225
Assets	92.267	-	91.345	-	71.325	88.377
Cash flow/assets	0.334	-	0.298	-	0.141	0.096
Equity/assets	0	-	0	-	0.052	0
$\lambda = 0$						
Fraction	0.137	0	0.075	0	0.147	0.641
Investment/assets	-0.022	-	-0.012	-	0.301	0.143
Debt/assets	-0.073	-	-0.056	-	0.254	0.143
Tobin's Q	1.721	-	1.643	-	3.095	2.505
Assets	89.456	-	85.385	-	72.753	76.768
Cash flow/assets	0.359	-	0.306	-	0.157	0.114
Equity/assets	0	-	0	-	0.059	0

5.6 Table IV: financing regimes and comparative statics

Read. Table IV sorts simulated observations by financing regime and compares the baseline estimate $\lambda = 0.059$ with a counterfactual $\lambda = 0$.

In the baseline panel:

- firms are net savers with distributions **19.9%** of the time,
- net savers with neither distributions nor equity issuance **8.6%** of the time,

- debtors issuing equity **9.1%** of the time,
- debtors in the interior zero-distribution regime **62.4%** of the time.

Equity issuers have the highest leverage, the highest investment rate, and the highest Q :

- $\text{debt/assets} = 0.286$,
- $\text{investment/assets} = 0.292$,
- Tobin's $Q = 2.846$.

Distribution-making savers have the lowest investment and relatively low Q :

- $\text{debt/assets} = -0.086$,
- $\text{investment/assets} = -0.032$,
- Tobin's $Q = 1.945$.

When λ is set to zero, the broad regime patterns remain similar, but equity issuance rises and saving falls.

Economic message.

- High- Q firms tend to be the firms issuing equity.
- Because debt adjusts sluggishly, that same pattern generates a negative relation between current leverage and lagged weighted Q .
- The result survives even when $\lambda = 0$, so the tax environment matters at least as much as the flotation-cost channel.

5.7 Counterfactual tax experiments

Read. The paper reports three especially revealing tax counterfactuals:

- When the maximum corporate tax rate is lowered to just above τ_i , the firm finances solely with equity about 3% of the time.
- When the maximum corporate tax rate is lowered below τ_i , the firm always retains funds and finances only with equity.
- When the maximum corporate tax rate is raised to 40%, some firms optimally issue debt and make distributions at the same time.

Economic message. These experiments show that the trade-off model is much richer dynamically than the static pecking-order rules of thumb suggest. Taxes can rationalize both corporate liquidity holdings and simultaneous debt issuance plus distributions.

6 Short summary

The paper builds a dynamic trade-off model in which firms choose leverage, saving, payouts, and real investment jointly under taxes, distress costs, collateral constraints, and equity flotation costs. The crucial insight is that the value of debt depends on the firm's current and anticipated financing margin. A marginal dollar of debt is more attractive when it replaces external equity than when it merely finances a higher payout. This simple regime logic breaks the static link between profitability and leverage and implies path dependence and hysteresis in leverage.

On the empirical side, the authors solve and simulate the model, then estimate its structural parameters with indirect inference on Compustat data. The model reproduces the main leverage facts usually seen as problems for the trade-off theory, including negative relations of leverage with cash flow, EBITDA, and weighted Q . The estimate of the cost of external equity is modest, around 5.9%, suggesting that large informational frictions are not needed to explain the data.

7 Conclusion

Core conclusion. The paper argues that the most famous empirical “failures” of the static trade-off theory are not failures of tax-based optimizing behavior. They are failures of using a static theory to interpret data generated by dynamic firms.

Economic intuition. Once firms can save, distribute, issue equity, and invest in a forward-looking way, the debt decision becomes regime-dependent. The relevant question is not simply “is debt tax-favored?” but rather “what does one more dollar of debt replace today, and how does it change tomorrow's financing margin?”

What the paper explains.

- profitable firms can borrow less,
- firms can hold cash even when debt has tax advantages,
- leverage can depend on lagged leverage,
- leverage can be negatively related to weighted Q without market timing,
- modest flotation costs are enough once taxes and dynamic margins are modeled properly.

Why the paper matters. This paper is influential because it shows how to connect corporate-finance theory to empirical patterns using a structural dynamic model rather than a static benchmark. It also sharpens the distinction between *reduced-form signs* and *structural interpretation*: the same negative leverage–profitability coefficient can come from pecking order, market timing, or a dynamic trade-off model.

Bottom line. The right lesson is not that trade-off theory fails. The right lesson is that leverage is a dynamic state variable, and the tax benefit of debt must be evaluated at the correct financing margin.